Searching the Cognitive Possibility Space:
A Computational Framework for Discovering Learnable Capabilities

Noam Shalom
Reichman University
noamshalom77@gmail.com

October 2026, working paper

Abstract

Reading is a capability evolution never selected for: culture found an input system that the human brain can learn so well that it becomes automatic. This raises a question that is usually not asked directly: *what else could a fixed learner become, given the right interface and training environment?* We formalise this as **Cognitive Capability Search (CCS)**: an outer loop that searches over interfaces, objectives, input conditions and plasticity schedules for a fixed-architecture learner, scored by how much new world structure becomes cheaply readable (*capability gain*) minus damage to existing abilities (*interference*), without naming a target capability. In a deliberately small toy (a modular neural network that "develops" with access to only half of its world's latent causes), CCS beats random search under an equal budget in 10 of 10 worlds ($p = 0.002$), and the environments it selects raise few-shot performance on a held-out battery of tasks that depend on previously inaccessible latents from 0.28 to 0.51–0.60 R^2. The search's most useful finding was one we did not design in: *acquire the new channel while the old sense is silent*. Changing only this one setting moves interference from ≈ 0.01 to 0.35–0.97. A revert test shows why: with the old sense silent, its pathway receives no gradient, so the search rediscovered gradient isolation by choosing an input regime. Region-level "adaptive plasticity" did not beat global plasticity, and the search exploited a gap in our first interference metric; closing the gap and replicating in ten worlds left every finding standing. We report these as what a toy can show: a framing and a working method, not a claim about human brains.

1 Motivation: reading as an existence proof

Writing is roughly five thousand years old, far too recent for the brain to have evolved for it. Yet literate adults read automatically, and reading is supported by a reproducible cortical territory in the ventral visual stream. The *neuronal recycling* hypothesis explains this by saying that cultural inventions succeed when they fit pre-existing circuitry closely enough to repurpose it [1]. Sensory substitution and augmentation make the same point from the other side: tactile images can support visual-like perception [2], a vibrating belt can become a usable sense of magnetic north [3], rats can learn to perceive infrared light [4] or navigate by a geomagnetic prosthesis [5], and skin can carry far more information than everyday life asks of it [6].

What these cases share is that *someone guessed the capability first*, then built the interface. Brain–machine interface research already treats the human and the decoder as two co-adapting learners [7, 8], but the task is still fixed in advance. This paper asks a different question:

Can we systematically search for capabilities a learner is able to acquire, rather than deciding the target capability in advance?

In other words, not "how do I teach it X?" but "*what can this learner become?*"

2 Cognitive Capability Search

Ingredients. A learner L with fixed architecture and partly developed weights; a space E of environments; and three measurements taken after L is exposed to $e \in E$:

- **Capability gain** $G(e)$: how much more of the world's latent structure becomes *cheaply* readable from L's internal representation (few-shot linear probes [9]), relative to before.
- **Interference** $I(e)$: loss of performance on abilities L already had, read through their original, frozen readouts.
- **Cost** $T(e)$: how long the learner had to be plastic.

CCS solves

$$e^\star = \arg\max_{e \in E} \; G(e) - \beta\, I(e) - \gamma\, T(e), \tag{1}$$

with no term that names a particular capability. The probe bank is generic (it asks about the world's generative factors, not about a task we hope to see), and a separate held-out task battery, never used by the search, checks whether what was found is useful.

Two loops. The *inner loop* is ordinary learning: L is trained on e's self-supervised objective. The *outer loop* searches over e. Because the outer loop can also choose *which parts of L are plastic and when*, a single environment can implement an *unlock* $\rightarrow$ *learn* $\rightarrow$ *select* $\rightarrow$ *consolidate* cycle, a computational analogue of controlled critical periods [10].

Relation to existing work. CCS borrows its machinery from open-ended search and environment design (POET [11], PAIRED [12], AI-generating algorithms [13]), from novelty and quality-diversity search [14, 15], from intrinsic motivation [16], curriculum learning [17] and meta-learning [18]. The difference is the object of the search. Those methods mostly search for environments that make an *agent* better at a given objective, or generate ever-harder tasks. CCS fixes the learner and asks which *new capabilities* are reachable for it, and at what price in interference with what it already does. This makes the stability–plasticity problem [19, 20] as a constraint on the search space rather than as a nuisance.

3 A toy instantiation

World. Eight latent causes $z \sim \mathcal{N}(0, I_8)$. The learner's innate sense $s \in \mathbb{R}^{16}$ is a fixed nonlinear mixture of $z_0..z_3$ only; $z_4..z_7$ are invisible to it.

Learner. A small modular network (all code is pure NumPy): $h_1 = \tanh(W_1 s + b_1)$ (sensory region R_1), $h_c = \tanh(W_c c)$ (interface region R_c), $h_2 = \tanh(W_{2a} h_1 + W_{2b} h_c + b_2 + \eta)$ (association region R_2, intrinsic noise $\eta \sim \mathcal{N}(0, 0.2^2)$), and a readout R_3 predicting $z_0..z_3$. "Development" trains R_1, R_2, R_3 on the innate task with the new channel silent (innate $\mathrm{R}^2 = 0.968$). The pathway from R_c into R_2 starts weak ($0.15\times$ normal scale), so at birth a new channel reaches the association region only below the noise floor.

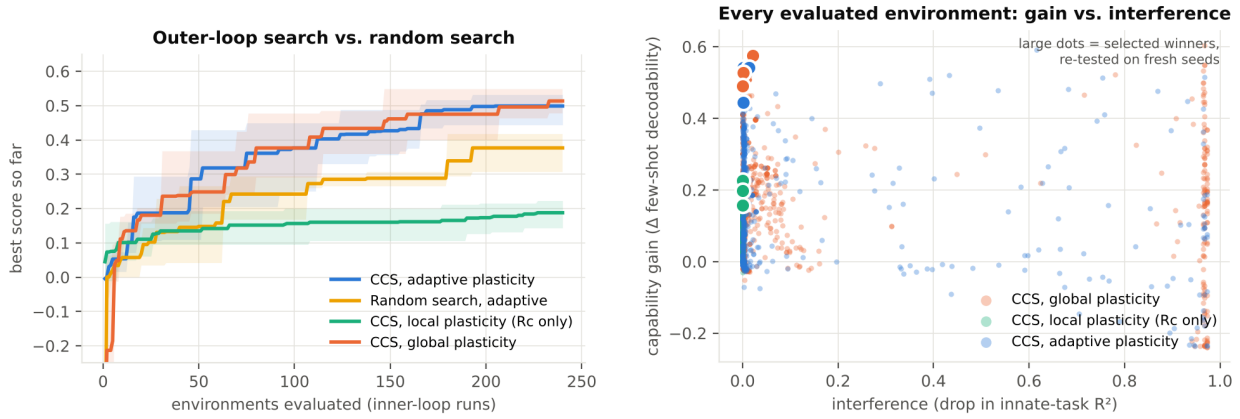


Figure 1: Left: best score found vs. number of inner-loop runs (mean over 3 worlds; band = min–max). **Right:** every environment evaluated by the three CCS searches. Most of the space trades capability for interference; the winners (large dots, re-tested on fresh seeds) sit in the top-left corner.

Environment space E. Each environment is a genome: *interface*: which latents a new 16-d channel carries (any subset of 8), its code (`linear`, Gaussian-tuned `population`, high-frequency `phase`, or `hashed` lookup) and noise level; *objective*: weights on reconstructing the channel from h_2 and on *aligning* channel-only h_2 to the representation the innate sense produces (a direct "recycling" signal); *input condition* during learning: channel only, both, or mixed; *plasticity schedule*: two phases, each unlocking any subset of $\{R_1, R_c, R_2\}$ for 0–600 steps; and learning rate. That is about 1.5×10^8 distinct environments.

Measurements. G is the mean change in few-shot (n=64) ridge-probe R^2 for each of the 8 latents from h_2, under two read-out conditions (channel only; sense + channel). I is the drop in innate-task R^2 through the frozen readout R_3, channel silent. β=1, γ=0.02 per 1200 steps.

Outer loop and baselines. A genetic algorithm (population 20, 12 generations, tournament selection, crossover, mutation, 2 elites) gives 240 inner-loop runs per search. We compare three plasticity regimes, *global* (all regions open for 600 steps), *local* (only R_c), and *adaptive* (schedule searched), and random search over the adaptive space with the same budget. Two hand-designed environments represent "the obvious first guess": carry all latents in a clean linear code, train by reconstruction with both inputs present, under global or local plasticity. Every result is repeated in three independently generated worlds (different mixing, codes and learner initialisation); winners are re-tested with three fresh inner-loop seeds. The full pipeline runs in about 15 minutes on two CPU cores.

4 Results

Search beats random sampling. With 240 inner-loop runs, GA search reached a best score of 0.50 (adaptive) and 0.51 (global) against 0.38 for random search (Fig. 1, left). Across 720 random environments only 3 (0.4%) exceed 0.3; the GA crossed 0.3 after 30–114 evaluations, random search after 137–192. Re-testing winners on fresh seeds reproduced their scores (Table 1), so this is not a winner's-curse artefact.

The capability that emerges. Fig. 2 shows one selected environment. Before exposure, the learner's association region carries essentially nothing about $z_4..z_7$. After 150 steps in the selected environment,

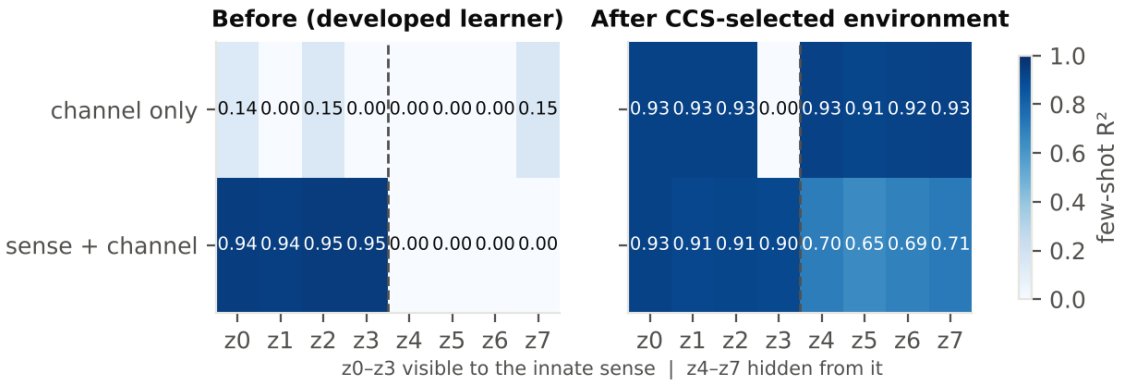


Figure 2: Few-shot decodability of each latent from the association region, before and after the CCS-selected environment (adaptive regime, world 0: carries $z_{0,1,2,4,5,6,7}$, linear code, reconstruction objective, channel-only input, $R_c{+}R_2$ open for 150 steps).

	search objective (fresh seeds)			held-out battery, R^2		
Condition	score	gain G	interf. I	all	visible	hidden
No interface	n/a	n/a	n/a	0.379	0.829	0.283
Hand-designed, global	-0.183	0.661	0.834	0.740	0.785	0.730
Hand-designed, local	0.121	0.131	0.000	0.460	0.819	0.384
Random search, adaptive	0.378	0.390	0.002	0.514	0.796	0.454
CCS, global plasticity	**0.513**	0.531	0.008	0.636	0.792	0.603
CCS, local plasticity	0.184	0.194	0.000	0.403	0.825	0.314
CCS, adaptive plasticity	0.499	0.509	0.005	0.620	0.795	0.583

Table 1: Means over 3 worlds $\times$ 3 inner-loop seeds. The battery is 40 random nonlinear functions of 2–3 latents, read out few-shot from h_2 with sense and channel present; "hidden" tasks depend on at least one of $z_4..z_7$ (33 of 40).

every carried latent is readable from the channel alone ($R^2 \approx 0.92$), and with sense and channel together the hidden latents reach $R^2 \approx 0.65$–0.71 while the visible ones stay at 0.90–0.93. Across all three worlds, the search consistently chose clean `linear` codes carrying 5–7 latents and avoided `phase` and `hashed` codes, which on average *reduced* decodability (Fig. 4). That interfaces differ sharply in learnability is the toy analogue of the recycling claim that only some cultural codes "fit".

Transfer to tasks the search never saw. On the held-out battery (Table 1, Fig. 3), CCS-selected environments roughly double performance on tasks that need a hidden latent ($0.28 \rightarrow 0.58$–0.60) at a small cost on tasks over visible latents ($0.83 \rightarrow 0.79$). The hand-designed global environment does better still on the battery (0.73), but it does so by destroying the learner's existing ability: innate-task R^2 falls from 0.97 to 0.11. The battery fits fresh readouts and so cannot see this; the interference term can. The two numbers have to be read together.

What the search found that we did not design: learn while the old sense is silent. All twelve winners chose the *channel-only* input condition. The CCS global winners differ from the hand-designed global environment in small ways (one latent fewer, sometimes more noise or a lower learning rate) and in one large way: they present the new channel *without* the innate sense. Table 3 isolates this: holding everything else fixed and changing only the input condition keeps the gain (0.51–0.57) but moves interference from

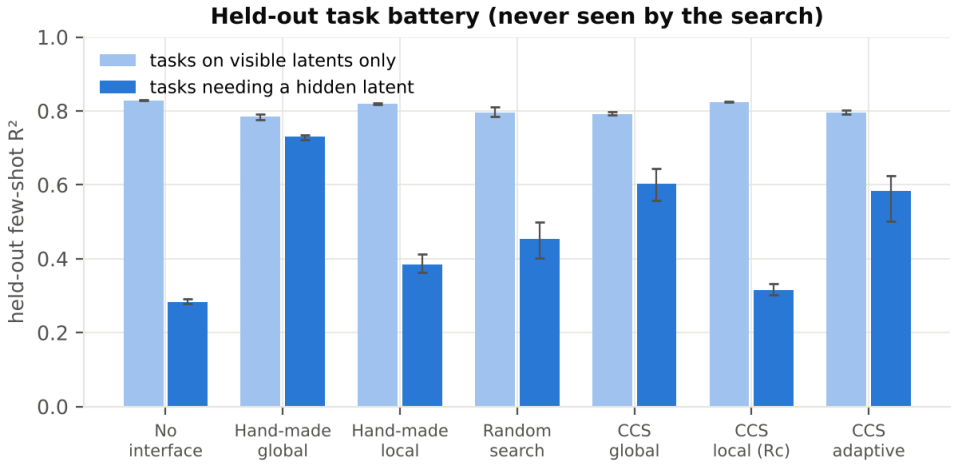


Figure 3: Held-out battery, mean over 3 worlds (error bars = min–max).

	innate R^2	recovered by reverting		Δpre-activation (signal SD)		$\|\Delta W_1\|/\|W_1\|$
input during learning	after	W_1	W_{2a}	constant	input-dependent	
channel only (as found)	0.960	0.000	0.008	0.01	0.10	0.00
both	0.068	0.827	0.443	0.02	0.79	0.23

Table 2: Mechanism of protection, CCS global winners, mean over 3 worlds. "Recovered by reverting": gain in innate R^2 when that one parameter group is reset to its developed value after acquisition. Reverting b_1, b_2, W_c or W_{2b} changes innate R^2 by < 0.001 in both conditions.

≤ 0.01 to 0.35–0.97.

Mechanism: input gating is gradient gating. Two measurements explain it (Table 2). With $s = 0$, the sensory weights W_1 receive exactly zero gradient and h_1 is a constant vector, so the gradient on W_{2a} is low-rank and its effect on real sensory input is small: the input-dependent part of the change to R_2's pre-activation is 0.10 signal-SD, against 0.79 when the sense is present. A *revert test* confirms where the damage lives: after training with both inputs, restoring W_1 alone to its developed value recovers 0.83 of innate R^2 and restoring W_{2a} recovers 0.44; after channel-only training there is nothing to recover. All adaptation went into the new pathway (W_c, W_{2b}). In other words the search rediscovered *gradient isolation*, the principle behind weight freezing, progressive networks and related continual-learning methods [20], by choosing an input regime, without being told anything about the architecture, while the "obvious" hand-designed regime missed it. That is both the argument for CCS and the limit of the claim: in hindsight the principle is familiar; what the search contributes is finding it. The pattern is loosely reminiscent of the recruitment of deprived cortex by Braille in blind readers [21], but we make no claim beyond the toy.

Adaptive plasticity: a negative result, with a caveat. We expected region-level control ("open R_2 briefly, then freeze") to beat global plasticity. It did not: 0.499 vs. 0.513 here, and 0.395 vs. 0.409 in the ten-world replication below ($p = 0.75$). Once input gating was available, it did the work that plasticity gating would otherwise do. The caveat is in Table 3: when the protective input condition is removed, the adaptive winners' shorter, partial schedules keep interference at 0.35 against 0.83 for global plasticity. Plasticity control looks like a second line of defence rather than the main one; this is suggestive only, with three worlds. Purely

Winner	input during learning	gain	interference	score
CCS, global	channel only (as found)	0.531	0.008	**0.513**
	mixed	0.571	0.968	−0.407
	both	0.553	0.827	−0.284
CCS, adaptive	channel only (as found)	0.509	0.005	**0.499**
	mixed	0.546	0.763	−0.221
	both	0.510	0.350	0.157

Table 3: Ablation: only the input condition is changed. Means over 3 worlds × 3 seeds.

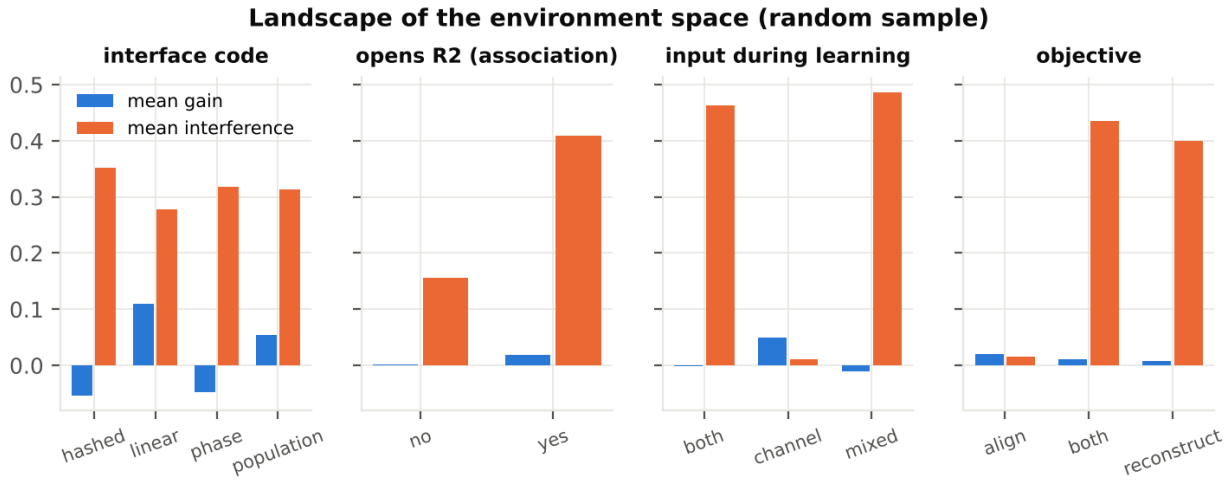


Figure 4: Mean gain and interference per gene value, over 720 random environments (an unbiased sample of E).

local plasticity (only the interface region) was the clearest failure: R_c alone cannot amplify a pathway into R_2 that starts below the noise floor, so gains plateaued at ≈ 0.19. A new capability needed the association region to change.

The search exploited our metric. Interference was measured with the new channel silent. With the channel switched *on*, CCS winners' innate-task R^2 drops from 0.97 to 0.88 (vs. 0.31 for the hand-designed global environment and 0.96 for local plasticity). Nothing in the score penalised this crosstalk, so the search did not avoid it. This is the general lesson of any objective-driven search, and for CCS it is a design requirement: the interference term must cover every context in which old abilities will be used.

Replication with the gap closed. We therefore re-ran the study in ten independently generated worlds with a corrected metric ("v2"): interference is the *worse* of the innate task's drop with the channel silent and with it on. Table 4 summarises; the original three worlds are included. Every finding survives. CCS beats random search in all ten worlds under the adaptive regime (sign test $p = 0.002$) and nine of ten under the global regime ($p = 0.021$), by 0.11–0.13 score. Across the 2,400 random environments sampled in this run only 0.3% score above 0.3; the adaptive search reaches 0.3 in nine of ten worlds (median 99 evaluations), random search in four. All twenty winners again chose channel-only input and a linear code, carrying 7 of the 8 latents on average. Adaptive and global plasticity remain indistinguishable ($\Delta = -0.015 \pm 0.055$, adaptive better in 4 of 10 worlds). What the corrected metric changed is informative in itself: the search did *not* find an environment free of crosstalk. The winners' innate R^2 with the channel on stays at 0.89–0.90,

Condition	score (mean $\pm$ 95% CI)	gain	interference	innate, channel on	battery, hidden
Hand-designed, global	-0.212 ± 0.047	0.658	0.860	0.295	0.726
Hand-designed, local	0.111 ± 0.014	0.127	0.006	0.961	0.380
Random search, adaptive	0.285 ± 0.024	0.357	0.065	0.903	0.401
CCS, global plasticity	$\mathbf{0.409 \pm 0.041}$	0.501	0.081	0.886	0.525
CCS, adaptive plasticity	$\mathbf{0.395 \pm 0.028}$	0.468	0.067	0.900	0.506

Table 4: Replication: 10 worlds, interference metric v2 (penalises crosstalk), winners re-tested on 3 fresh seeds each. CI over worlds.

so they now pay the penalty (interference 0.07–0.08, almost all of it crosstalk) for essentially the same gain (on the same ten worlds, 0.47–0.50 under v2 against 0.49–0.50 under v1). Within this environment space, some crosstalk appears to be the price of a new channel entering a shared association region at all. Under the original metric on the same ten worlds the scores are 0.488 ± 0.042 (global) and 0.476 ± 0.043 (adaptive), so the three-world numbers of Table 1 were not a lucky draw.

5 What this does and does not show

It shows that, for a fixed learner, the space of interfaces and training regimes can be searched *for capabilities* rather than *for a task*; that such search finds far better regions of that space than random sampling; that what it finds transfers to tasks it never saw; and that it can surface non-obvious training principles (here, input gating) that a designer can then test directly. It also shows the method's main hazard: it optimises exactly the interference you measure.

It does not show anything about human brains. The learner is a three-region $\tanh$ network; the "capability" is linear decodability of a synthetic world's factors; ten worlds and 240 evaluations per search are small, and the environment space is one we designed. The probe bank, though generic, is still a choice: capabilities that are not functions of z (new kinds of computation rather than new information) are invisible to it.

6 Why this might matter

The near-term version of CCS is practical: searching the design space of sensory-substitution devices, BCIs and training curricula with an explicit interference budget, using models of the learner fitted to behavioural or neural data as the inner loop. The far version is a question: are there cognitive modalities with no cultural analogue yet (for example, a brain that has learned to treat another person's neural state as an ordinary perceptual input) sitting inside the constraints of the human brain, undiscovered because nobody has built the right interface or training environment? We do not claim this toy says anything about that. We claim it is the right shape of question, and that it can be posed as a search.

Reproducibility. Code, results and figures are released (NumPy and Matplotlib only); `run_experiments.py`, `run_ablation.py`, `run_mechanism.py` and `make_figures.py` regenerate every number here, and `-metric v2 -worlds 10` reproduces the replication.

How this was made. The research question, the framing of capability search and the decisions about what to test are the author's. Code, experiments, figures and the draft were produced with Claude (Anthropic) working from those decisions; every number is generated by the released code and was re-run in a clean

environment. The replication and the mechanism analysis were added after the first version to test the main finding rather than to extend it.

References

[1] Stanislas Dehaene and Laurent Cohen. Cultural recycling of cortical maps. *Neuron*, 56(2):384–398, 2007.

[2] Paul Bach-y Rita, Carter C. Collins, Frank A. Saunders, Benjamin White, and Lawrence Scadden. Vision substitution by tactile image projection. *Nature*, 221(5184):963–964, 1969.

[3] Saskia K. Nagel, Christine Carl, Tobias Kringe, Robert Märtin, and Peter König. Beyond sensory substitution—learning the sixth sense. *Journal of Neural Engineering*, 2(4):R13–R26, 2005.

[4] Eric E. Thomson, Rafael Carra, and Miguel A. L. Nicolelis. Perceiving invisible light through a somatosensory cortical prosthesis. *Nature Communications*, 4:1482, 2013.

[5] Hiroaki Norimoto and Yuji Ikegaya. Visual cortical prosthesis with a geomagnetic compass restores spatial navigation in blind rats. *Current Biology*, 25(8):1091–1095, 2015.

[6] Scott D. Novich and David M. Eagleman. Using space and time to encode vibrotactile information: Toward an estimate of the skin's achievable throughput. *Experimental Brain Research*, 233(10):2777–2788, 2015.

[7] Krishna V. Shenoy and Jose M. Carmena. Combining decoder design and neural adaptation in brain-machine interfaces. *Neuron*, 84(4):665–680, 2014.

[8] Serafeim Perdikis and José del R. Millán. Brain-machine interfaces: A tale of two learners. *IEEE Systems, Man, and Cybernetics Magazine*, 6(3):12–19, 2020.

[9] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. arXiv:1610.01644, 2016.

[10] Takao K. Hensch. Critical period plasticity in local cortical circuits. *Nature Reviews Neuroscience*, 6(11):877–888, 2005.

[11] Rui Wang, Joel Lehman, Jeff Clune, and Kenneth O. Stanley. Paired open-ended trailblazer (POET): Endlessly generating increasingly complex and diverse learning environments and their solutions. arXiv:1901.01753, 2019.

[12] Michael Dennis, Natasha Jaques, Eugene Vinitsky, Alexandre Bayen, Stuart Russell, Andrew Critch, and Sergey Levine. Emergent complexity and zero-shot transfer via unsupervised environment design. In *Advances in Neural Information Processing Systems*, volume 33, pages 13049–13061, 2020.

[13] Jeff Clune. AI-GAs: AI-generating algorithms, an alternate paradigm for producing general artificial intelligence. arXiv:1905.10985, 2019.

[14] Joel Lehman and Kenneth O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19(2):189–223, 2011.

[15] Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. arXiv:1504.04909, 2015.

[16] Pierre-Yves Oudeyer, Frédéric Kaplan, and Verena V. Hafner. Intrinsic motivation systems for autonomous mental development. *IEEE Transactions on Evolutionary Computation*, 11(2):265–286, 2007.

[17] Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th International Conference on Machine Learning*, pages 41–48, 2009.

[18] Timothy Hospedales, Antreas Antoniou, Paul Micaelli, and Amos Storkey. Meta-learning in neural networks: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(9):5149–5169, 2022.

[19] Michael McCloskey and Neal J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of Learning and Motivation*, volume 24, pages 109–165. Academic Press, 1989.

[20] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017.

[21] Norihiro Sadato, Alvaro Pascual-Leone, Jordan Grafman, Vicente Ibañez, Marie-Pierre Deiber, George Dold, and Mark Hallett. Activation of the primary visual cortex by Braille reading in blind subjects. *Nature*, 380 (6574):526–528, 1996.